

MSc Creative Computing

Practice as Research — Lecture 4 **Choosing and designing methods**

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Practice as Research

Choosing and Designing Methods

Lecture 4 of 8 — Practice as Research

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Recap: where we are

Week 1: Research = edge + hypothesis + test.

Week 2: Practice produces knowledge — procedural, experiential, tacit, propositional, critical.

Week 3: Finding the gap — situating your work in a field.

Today: **How do you investigate your gap?**

A gap tells you what's missing. A method tells you how you're going to fill it — and why that approach is appropriate.

What is a method?

A method is a **systematic way of doing something that can be described, justified, and repeated** (at least in principle).

It is not the same as a tool. Python is not a method. TensorFlow is not a method. A method is the *logic* behind what you do with those tools.

"I trained a neural network" is not a method.

"I trained a neural network on X data, varied Y parameter, measured Z outcome, and compared against baseline W" — that's closer.

Methods are not one-size-fits-all

Different research questions need different methods. The method has to match:

The type of gap — a knowledge gap needs investigation; a method gap needs construction and evaluation; a critical gap needs analysis and argumentation.

The type of knowledge you want to produce — propositional knowledge needs controlled comparison; experiential knowledge needs encounter and documentation; tacit knowledge needs reflective practice.

The practical constraints — time, access, equipment, ethics, your own skills.

There is no single correct method for any project. There are better and worse fits.

Haseman's three paradigms

Haseman (2006) argues that practice-led research doesn't fit neatly into the two established paradigms (Haseman, B. (2006). A Manifesto for Performative Research. Media International Australia, 118(1), 98-106.):

Quantitative — hypothesis testing, controlled variables, statistical analysis. Produces numerical data. Answers "how much" and "how often."

Qualitative — interviews, ethnography, thematic analysis. Produces textual/descriptive data. Answers "what is it like" and "what does it mean."

Performative — Haseman's proposed third paradigm. The research output is not a number or a text — it is a **practice**, an artefact, a performance. The research is expressed *through* the doing.

Most creative computing projects involve elements of **all three**. The question is which is primary and how the others support it.

A toolkit of methods for practice-based research

Here are methods you are likely to use. You don't need all of them. You need the ones that fit your question.

Construction / system building — you make something. The making *is* the investigation. This is the core of most practice-based work in creative computing.

Technical evaluation — you measure your system's performance against a baseline or benchmark. Latency, accuracy, perceptual quality, computational cost.

User study / participant evaluation — you put your system in front of people and gather data on how they experience, use, or respond to it.

Reflective practice — you document your own process of making, noting decisions, surprises, dead ends, and discoveries. This is where tacit knowledge becomes (partially) explicit.

Comparative analysis — you compare your approach to existing approaches, technically or critically.

Critical/theoretical analysis — you use theory to interpret what your artefact does or reveals.

Construction as method

In creative computing, **building something is itself a research method** — not a preliminary to research.

This is what Borgdorff means by research *in and through* the arts, and what Smith & Dean's iterative cyclic web describes: the building process generates knowledge that couldn't have been produced any other way.

But construction alone is not sufficient. You also need to:

1. **Document** what you built and why — the design decisions, the architecture, the constraints.
2. **Contextualise** it — how does it relate to what already exists?
3. **Evaluate** it — does it do what you hoped? What did it reveal? What surprised you?

Construction + documentation + contextualisation + evaluation = a research contribution.

Technical evaluation: what to measure and how

If your project involves building a system, you will probably need some form of technical evaluation. Common approaches:

Benchmarking — compare your system's output to established baselines using standard metrics (e.g. FID for image generation, PESQ/POLQA for audio quality, latency measurements for real-time systems).

Ablation studies — remove or change components of your system to show what each part contributes.

Stress testing — push the system to its limits. Where does it break? What are the failure modes?

Reproducibility — can someone else run your code and get comparable results? Releasing code and data is part of the method.

Not all of these will apply to your project. Choose the ones that address your research question.

User studies and participant evaluation

If your project involves people experiencing, using, or responding to your work, you need a user study. Common approaches in creative computing:

Structured interviews — you ask participants predefined questions about their experience. Good for depth.

Questionnaires / surveys — standardised instruments (e.g. [SUS](#) for usability, [NASA-TLX](#) for cognitive load, [AttrakDiff](#) for user experience). Good for breadth and comparability.

Think-aloud protocols — participants narrate their experience while using your system. Good for understanding process.

Observation — you watch people interact with your work and note what happens. Good for behaviour you can't self-report.

Workshops / co-design sessions — participants contribute to the design process itself. Good for participatory and speculative projects.

Key resources:

- Lazar, Feng & Hochheiser (2017). [*Research Methods in Human-Computer Interaction*](#), 2nd ed. Morgan Kaufmann. — The standard HCI methods textbook.
- Cairns & Cox (eds.) (2008). [*Research Methods for Human-Computer Interaction*](#). Cambridge University Press.

Reflective practice as method

Reflective practice means **systematically documenting your own process of making** — not as a diary, but as evidence.

This draws on Donald Schön's (1983) concept of the **reflective practitioner**: someone who thinks about what they're doing *while they're doing it*, and uses that reflection to steer the next action.

Practical forms:

Studio/lab log — regular dated entries recording what you did, what happened, what you decided, and why. Digital or physical.

Annotated portfolio — a curated set of artefacts (prototypes, sketches, outputs, screenshots) with written commentary explaining what each reveals.

Process documentation — video, screen recordings, version histories, commit logs with meaningful messages.

The key: reflective practice is not "I did X." It is "I did X, expecting Y, but Z happened, which led me to revise my understanding of W."

Methods you can use without reinventing the wheel

You don't need to invent a new method. Established frameworks exist and are there to be used:

For building + evaluating interactive systems:

- [Research through Design \(RtD\)](#) — Zimmerman, Stolterman, and others. Design process as research inquiry. Well established in HCI.
- [Design Science Research](#) — Hevner et al. (2004). Build an artefact, evaluate it, articulate the contribution. Common in information systems.

For understanding user experience:

- [Thematic Analysis](#) — Braun & Clarke (2006). The standard method for analysing qualitative data (interviews, open-ended responses). Widely used, well documented, citable.
- [System Usability Scale \(SUS\)](#) — Brooke (1996). A 10-item questionnaire for quick usability assessment. Free, validated, takes 2 minutes to administer.

For documenting creative process:

- [Annotated Portfolios](#) — Gaver & Bowers (2012). A method for presenting design artefacts as research contributions.
- Nelson, R. (2013). [*Practice as Research in the Arts*](#). Palgrave Macmillan. — Chapter 3 on "multi-mode epistemological model" for PaR.

For technical comparison:

- Standard ML evaluation protocols — train/test splits, cross-validation, ablation. Any good ML textbook covers these.

Combining methods

Most practice-based projects in creative computing use a **mixed-methods** approach:

A typical structure might be:

1. **Construction** — build the system.
2. **Technical evaluation** — measure its performance.
3. **User study** — have people use it and gather their responses.
4. **Reflective practice** — document the design process and what you learned from it.
5. **Critical framing** — use theory to interpret the results.

You don't need all five. But you almost certainly need more than one.

The methods should be **integrated, not bolted on**. Each method should address a different aspect of the research question, and together they should produce a richer answer than any one method alone.

Writing a methods section

Your written submission will need a methods section. It should cover:

1. **What methods you used** — named and cited.
2. **Why you chose them** — how they match your research question and gap.
3. **How you applied them** — enough detail that someone could follow your process.
4. **What data they produced** — what kind of evidence you gathered.
5. **How you analysed that data** — the analytical framework.

A good methods section is specific and honest. It includes limitations: "I had 8 participants rather than 20 because of time constraints. This limits the generalisability of the findings but is sufficient for a qualitative exploration."

Ethics: why this matters

Any research that involves other people, their data, or potential risks requires **ethical approval before you begin**.

This is not bureaucracy for its own sake. It exists to protect:

- **Participants** — from harm, deception, or exploitation.
- **You** — from doing something that causes harm you didn't anticipate.
- **The institution** — from liability.
- **The field** — from losing public trust.

If your project involves human participants, secondary human data, external collaborators, or any of the categories we're about to discuss — you need to go through the ethics process. And you need to do it **before you collect any data**, not after.

CCI ethics procedures: what you need to do

All MSc students at CCI must complete the **ethics pre-screen checklist** before any data gathering begins.

The checklist is here: [CCI Ethics Pre-Screen Checklist](#)

Full details: [CCI Taught Course Ethics Procedures](#)

Your project is likely to have an ethical dimension if it involves:

- Other people — as participants, users, testers, or subjects of observation (including in a professional capacity)
- Human data — whether you collect it yourself (primary) or use existing datasets (secondary, e.g. social media posts, medical records)
- External collaborators — companies, organisations, community groups
- Animals, human tissue, or environmental impact
- Health and safety risks beyond everyday life

The most common scenario for this cohort: **you're building something and asking people to use it, or you're using a dataset that contains human-generated content.** Both require ethical consideration.

If the pre-screen says you need full approval

If the pre-screen identifies that your project has ethical dimensions requiring approval, you will need to:

1. Complete the [ethics application form](#) with your supervisor.
2. Prepare a [Participant Information & Consent Form](#).
3. Submit both to **cci-cresc@arts.ac.uk** with the subject line: **MSc project ethics application**.

Minimise risk. Most student projects should be **minimal risk** — defined as risk no greater than what participants encounter in everyday life. If your project goes above minimal risk (vulnerable participants, sensitive topics, deception, confidential records), it will be reviewed by the UAL educational ethics sub-committee, which takes longer.

If your project changes after approval, submit a [Notice of Amendment](#). Substantial changes may require a new application.

Questions? Contact **cci-cresc@arts.ac.uk** or **#research-ethics** on Slack.

Method design as research design

Choosing methods is not a box-ticking exercise. It is **research design** — the architecture of your investigation.

Good research design means:

- Your methods **match your question**. (Don't run a user study if your question is about computational efficiency.)
- Your methods **produce the right kind of evidence**. (Don't use a questionnaire if you need rich descriptions of experience.)
- Your methods are **feasible**. (Don't plan for 50 participants if you have 6 weeks.)
- Your methods are **ethical**. (Don't collect data you don't need. Don't expose participants to unnecessary risk.)
- Your methods are **documented and justified** in your writing.

The test: can you explain, in two sentences, what your method is and why it's the right one for your question?

An exercise: method matching

Here are three fictional project descriptions. For each, identify which methods from slide 6 would be appropriate, and why.

Project A: A real-time system that generates ambient soundscapes from weather data, intended for public installation.

Project B: A tool that helps non-programmers fine-tune image generation models, with the goal of making the process more accessible.

Project C: An artistic investigation of how language models represent gender, expressed as a series of generated text-and-image compositions.

Take 10 minutes. For each project, name 2–3 methods and write one sentence explaining why each is appropriate.

For next week

Week 5: **Documentation — what to document and how.**

You've now got a question, a gap, and a method. Next week we ask: what counts as evidence in practice-based research, and how do you capture it?

No required reading — but if you want to get ahead:

- Gaver, W. & Bowers, J. (2012). [Annotated Portfolios](#). *Interactions*, 19(4), 40–49.
- Keep a process log this week. Just notes — what you're doing, why, what's happening. We'll use these in Week 5.

References

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- Haseman, B. (2006). ["A Manifesto for Performative Research."](#) *Media International Australia*, 118(1), 98–106.
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 - Hevner, A. et al. (2004). ["Design Science in Information Systems Research."](#) *MIS Quarterly*, 28(1), 75–105.
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 - Brooke, J. (1996). "SUS: A 'Quick and Dirty' Usability Scale." In Jordan et al. (eds.), *Usability Evaluation in Industry*. Taylor & Francis, 189–194.
 - [CCI Taught Course Ethics Procedures](#)
 - [UAL Research Ethics Policy](#)

Questions

What method are you leaning towards?

What's stopping you from starting?

Does your project involve participants — and have you started the ethics process?